



College of Engineering
Mechatronics and mechanical engineering department
CCEE564-Machine Learning and Data Mining
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Project 1:

Machine Learning-Based Prediction of Power Output in Solar PV Systems

Report Submitted by:

Ahmad Achaji – 20210313

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Report Submitted to:

Ghalia Nassreddine

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Table of Contents

Introduction	3
Code discussion.....	3
EDA Results & Visualizations	3
Histogram Analysis	3
Box Plots	4
Scatter Plots & Correlation Matrix.....	5
Feature Selection & Comparison of Techniques.....	7
PCA	7
Chi-Square	7
Model Development & Hyper-Parameter Tuning	8
Performance Evaluation & Seasonal Influence Analysis.....	8
Conclusion	9

Table of Figures

Figure 1 Histogram of our data set	3
Figure 2 Box plot of the features.....	4
Figure 4 Scatter plot of our data set	5
Figure 5 Correlation Matrix of our data frame.....	6
Figure 6 PCA plot.....	7

Table of Tables

Table 1 Chi-square Table	8
Table 2 XGBoost and Random Forest Performance Evaluation.....	8

Introduction

The aim of this project is to create a machine learning module that can predict the output power of a solar system based on historical solar data. This data contains different data such as time, wind speed, sunshine, air pressure, radiation, air temperature, relative humidity and system production.

To achieve our goal we have many objectives, we are going to implement feature selection techniques, perform hyper-parameter tuning, analyze seasonal influence, and evaluate model performance.

Code discussion

EDA Results & Visualizations

Histogram Analysis

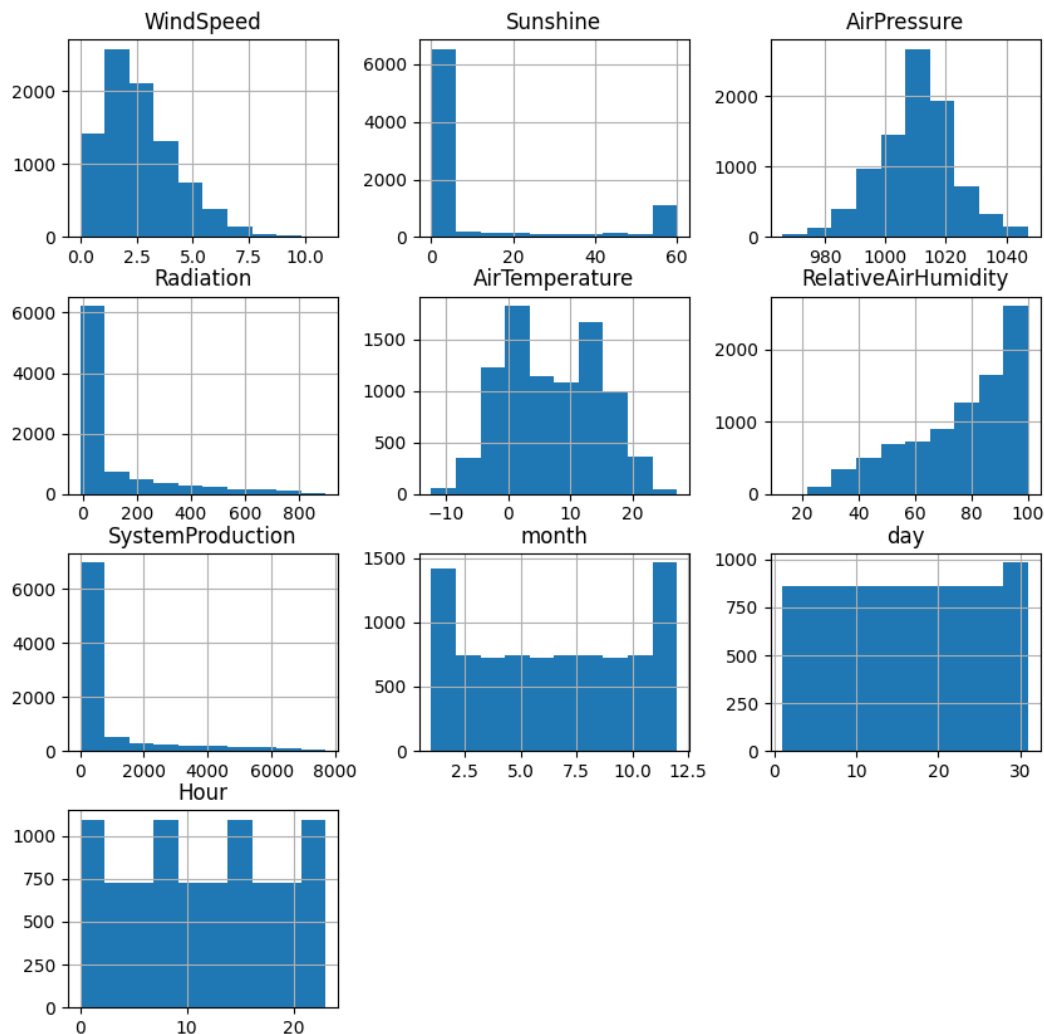


Figure 1 Histogram of our data set

Based on Figure 1 we can deduce that:

- The data is imbalanced and skew, that's because we can see that Radiation, Wind Speed, and System Production, exhibit highly skewed distributions. In other terms our data is imbalances; to solve this problem, we are going to apply a scaling method later.
- Sunshine, radiation and air temperature most probably have a strong relation with our target value system production (they have a string impact).

Box Plots

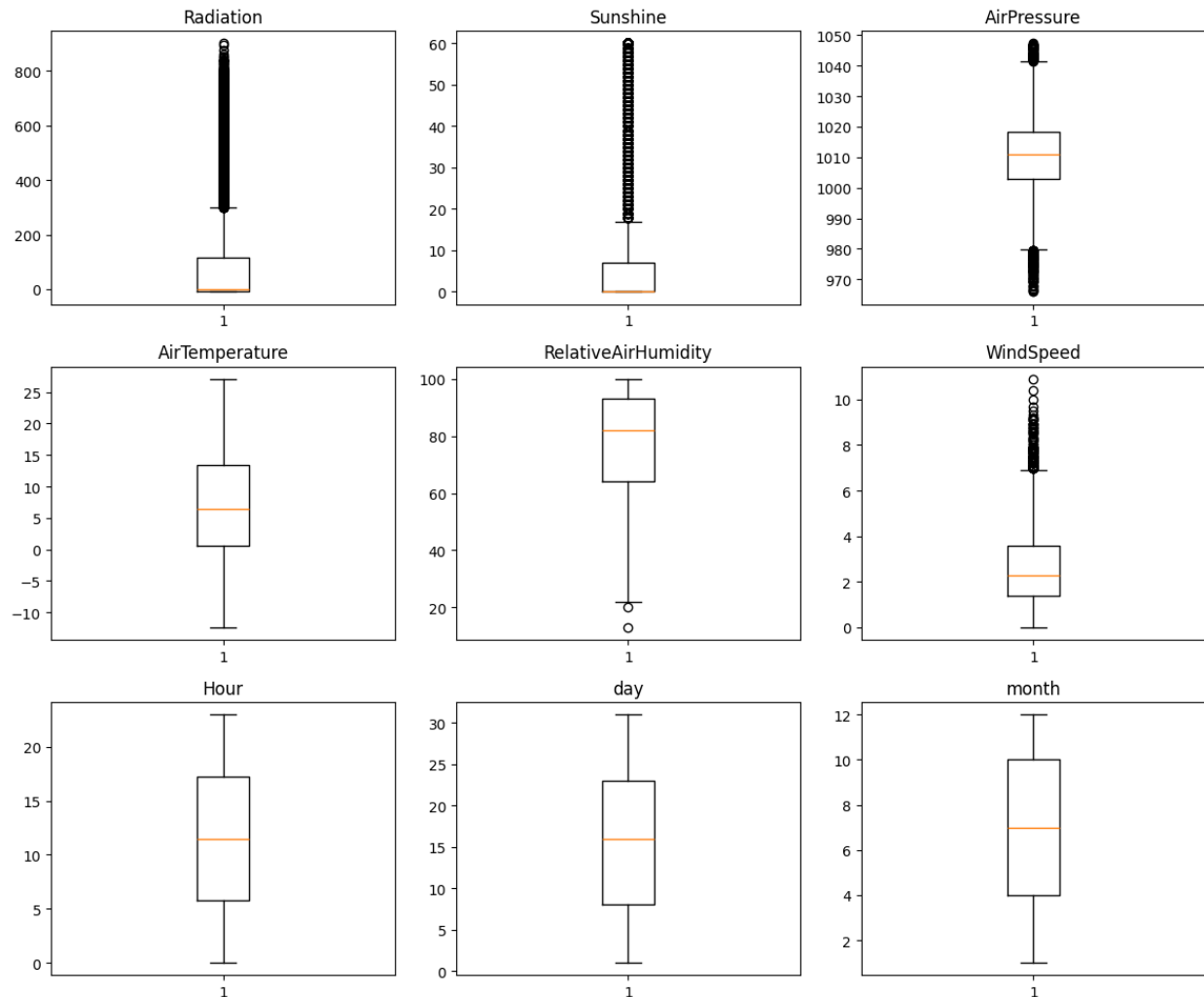


Figure 2 Box plot of the features

Based on figure 2 we can see that we have a lot of outliers in many features (e.g. Radiation, Sunshine, Windspeed, and Air pressure). To solve this problem, we are going to use a robust scale technique later.

Scatter Plots & Correlation Matrix

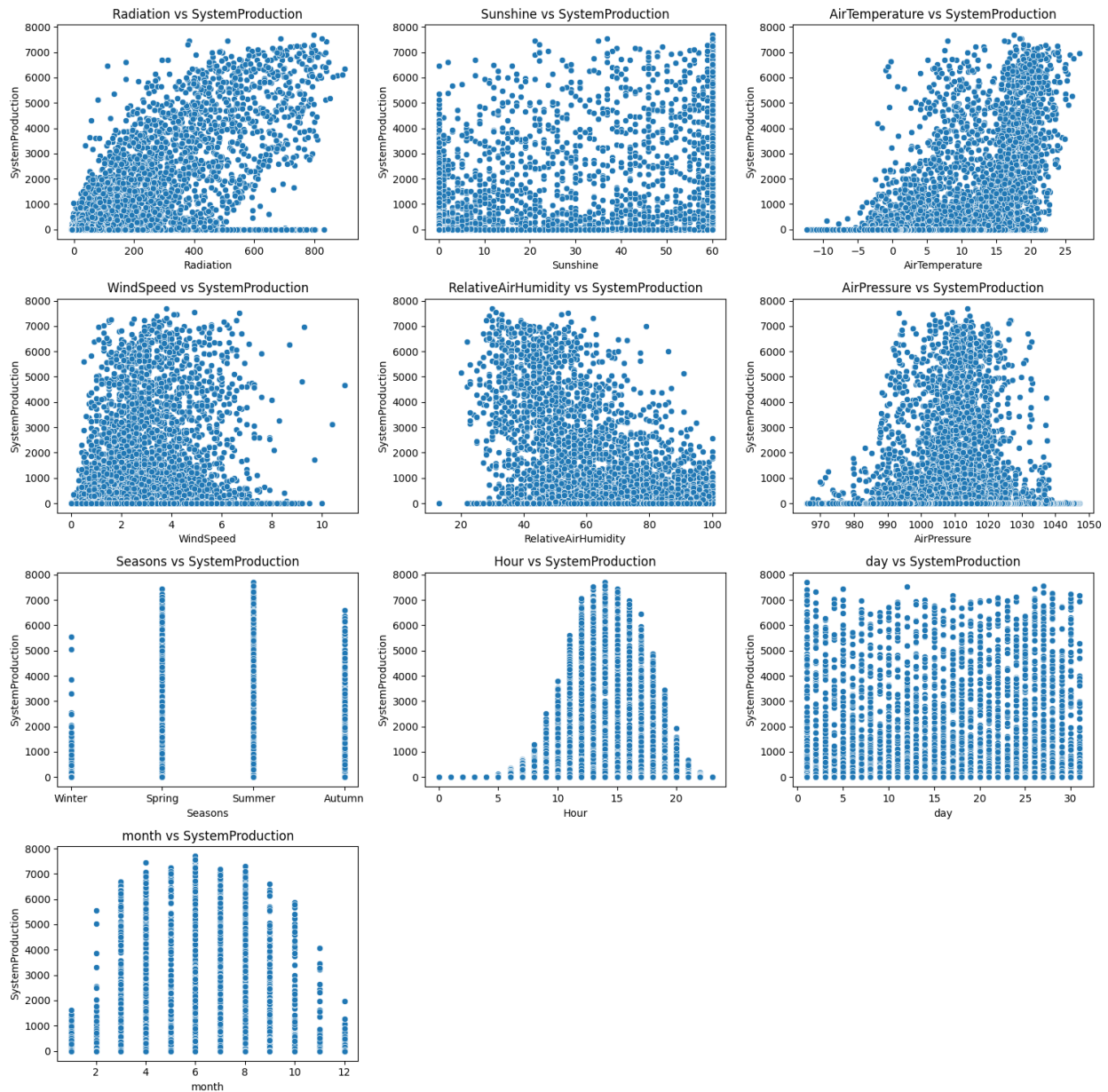


Figure 3 Scatter plot of our data set

We can deduce from figure 4 that we have a linear relationship between:

- Radiation and system production
- Air temperature and system production.

Spring and summer have the highest production rate among the seasons. That is translated also in the months where months 11, 12, 1, 2 have the lowest production, and 5-10 have the highest one.

If we check the hour scatter plot, we can see that the system production is at its maximum between 12 until 17 which is logical.

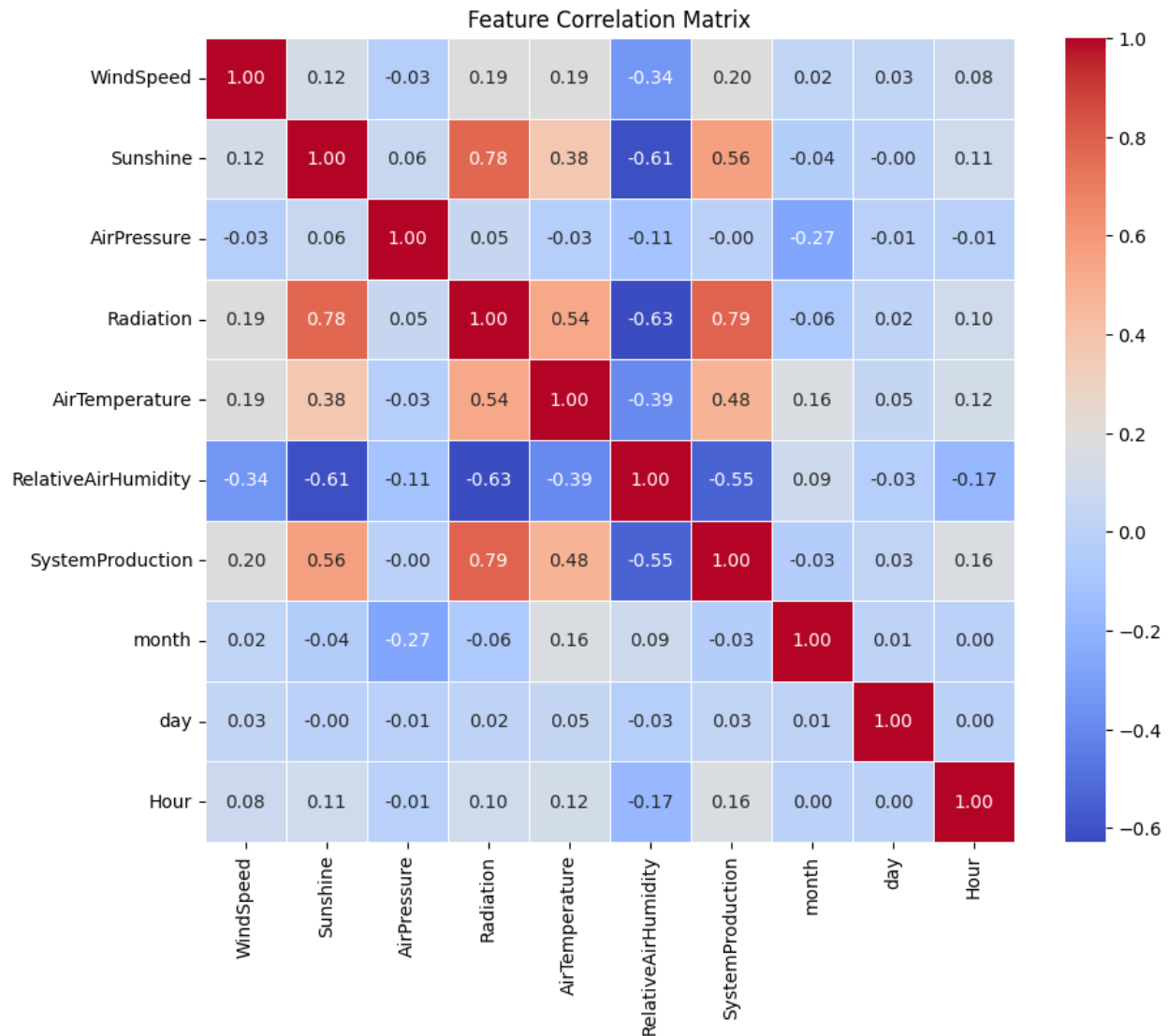


Figure 4 Correlation Matrix of our data frame

Based on figure 4 and 5 we can deduce that the system production is primarily driven by radiation, sunshine, and air temperature, while relative humidity negatively affects production. Features such as wind speed and hour have a small effect on the system production. All the other features have approximately a negligible effect on our target.

Moreover, we can see that production varies significantly across seasons, likely higher in summer and lower in winter.

Feature Selection & Comparison of Techniques

PCA

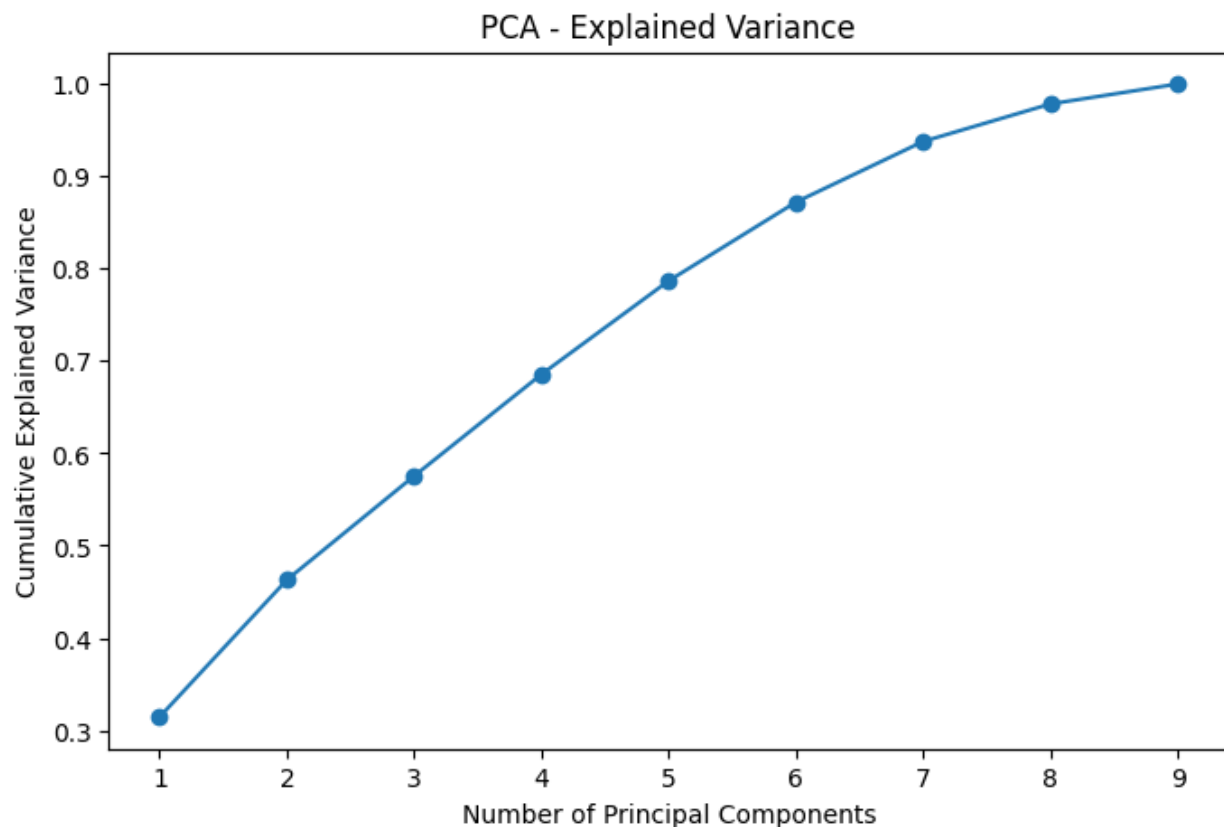


Figure 5 PCA plot

Figure 6 explains how much variance is captured as the number of principal components increases.

- first principal component: captures ~35% of the variance.
- 4th or 5th components capture 70-80% of the variance.
- 6th or 7th component: capture ~90% of the variance.

As conclusion, the key components range from 4 to 6 principal components, capturing around 75-90% of the variance. Using only these components effectively reduces data complexity while retaining most of the important information. However, reducing components further may lead to some loss of variability.

Chi-Square

This test is applied to assess the impact of categorical variables on SystemProduction. In our data set the only categorical variable for now is the seasons (we already dropped the “Date-Hour (NMT)” column).

Feature	Chi ² Score	P-Value
Seasons_Summer	695.700807	2.573926e-153
Seasons_Winter	535.144026	2.147080e-118
Seasons_Autumn	22.075201	2.621749e-06
Seasons_Spring	1.385401	2.391832e-01

Table 1 Chi-square Table

The results in table 1 indicate that Summer and Winter have the strongest impact due to their high Chi² scores and low p-values.

Model Development & Hyper-Parameter Tuning

We split our features and target into train test data, then we applied a Robust scaling method.

Two machine learning models are implemented:

1. XGBoost (Extreme Gradient Boosting):
Hyper-parameter tuning is performed using Grid Search with parameters: n_estimators, max_depth, and learning_rate.
Best parameters obtained: 'learning_rate': 0.1, 'max_depth': 7, 'n_estimators': 200}
2. Random Forest (RF):
Grid Search is used to optimize hyperparameters, including n_estimators and max_depth.
Best parameters obtained: 'max_depth': 15, 'n_estimators': 200}

Performance Evaluation & Seasonal Influence Analysis

The performance of both models is assessed using multiple metrics:

Here's your data formatted into a table:

Model	MAE	MSE	RMSE	R ²	NRMSE
XGBoost	198.44	212,265.82	460.72	0.8927	0.1072
Random Forest	208.82	221,667.58	470.82	0.8880	0.1120

Table 2 XGBoost and Random Forest Performance Evaluation

Table 2 Results show that the XGBoost model outperforms Random Forest in all key metrics, making it the preferred choice for predicting solar PV output. But still the difference between the two models in terms of precision isn't big, any model can be a good choice.

Conclusion

As conclusion we have achieved our aim by applying a machine learning model that can predict the PV power output, with a good efficiency (88-89% R^2). We have implemented feature selection techniques, performed hyper-parameter tuning, analyzed seasonal influence, and evaluated model performance. As summary we can see that the XG boost gives better result than the random forest, but still the two models give an acceptable result.